

M. Tech. Part I/PhD End Semester Examination, Session 2024 -25 Mining Engineering
Subject: Operations Research in Mining, Subject Code: MN561

Time: Three hours

Maximum Marks: 50

All the symbols have their usual meaning.

All question carry equal marks. **Answer any ten questions.**

1.	<i>Label each of the following statements about linear programming problems as true or false, and then justify your answer.</i> (a) For minimization problems, if the objective function evaluated at a CPF solution is no larger than its value at every adjacent CPF solution, then that solution is optimal. T (b) Only CPF solutions can be optimal, so the number of optimal solutions cannot exceed the number of CPF solutions. T (c) If multiple optimal solutions exist, then an optimal CPF solution may have an adjacent CPF solution that also is optimal (the same value of Z). T (d) If the value of the objective function is equal at two different feasible points x^* and x^{**}, then all points on the line segment connecting x^* and x^{**} are feasible and Z has the same value at all those points. T (e) If the problem has n variables (before augmenting), then the simultaneous solution of any set of n constraint boundary equations is a CPF solution.	[5]
2.	Write short notes on: (a) Characteristics of Dynamic programming problems (b) Basic structure of queuing model	[5]
3.	The Energetic Company needs to make plans for the energy systems for a new building. The energy needs in the building fall into three categories: (1) electricity, (2) heating water, and (3) heating space in the building. The daily requirements for these three categories (all measured in the same units) are Electricity: 20 units Water heating: 10 units Space heating: 30 units. The three possible sources of energy to meet these needs are electricity, natural gas, and a solar heating unit that can be installed on the roof. The size of the roof limits the largest possible solar heater to 30 units, but there is no limit to the electricity and natural gas available. Electricity needs can be met only by purchasing electricity (at a cost of \$50 per unit). Both other energy needs can be met by any source or combination of sources. The unit	[5]

	costs are																																																	
	<table><tr><td></td><td>Electricity</td><td>Natural</td><td>Solar</td></tr><tr><td>Water</td><td>\$90</td><td>\$60</td><td>\$30</td></tr><tr><td>heating</td><td>\$80</td><td>\$50</td><td>\$40</td></tr></table>		Electricity	Natural	Solar	Water	\$90	\$60	\$30	heating	\$80	\$50	\$40																																					
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	The objective is to minimize the total cost of meeting the energy needs. Formulate this problem as a transportation problem by constructing the appropriate parameter table and apply the transportation simplex method to obtain an optimal solution.																																																	
4.	In preparation for the winter season, a clothing company is manufacturing parka and goose overcoats, insulated pants, and gloves. All products are manufactured in four different departments: cutting, insulating, sewing, and packaging. The company has received firm orders for its products. The contract stipulates a penalty for undelivered items. Formulate a linear programming model to devise an optimal production plan for the company based on the following data: Time per unit (hr) <table><tr><td>Department</td><td>Parka</td><td>Goose</td><td>Pants</td><td>Gloves</td><td>Capacity</td></tr><tr><td>Cutting</td><td>.30</td><td>.30</td><td>.25</td><td>.15</td><td>1000</td></tr><tr><td>Insulating</td><td>.25</td><td>.35</td><td>.30</td><td>.10</td><td>1000</td></tr><tr><td>Sewing</td><td>.45</td><td>.50</td><td>.40</td><td>.22</td><td>1000</td></tr><tr><td>Packaging</td><td>.15</td><td>.15</td><td>.1</td><td>.05</td><td>1000</td></tr><tr><td>Demand</td><td>800</td><td>750</td><td>600</td><td>500</td><td></td></tr><tr><td>Unit profit</td><td>\$30</td><td>\$40</td><td>\$20</td><td>\$10</td><td></td></tr><tr><td>Unit penalty</td><td>\$15</td><td>\$20</td><td>\$10</td><td>\$8</td><td></td></tr></table>	Department	Parka	Goose	Pants	Gloves	Capacity	Cutting	.30	.30	.25	.15	1000	Insulating	.25	.35	.30	.10	1000	Sewing	.45	.50	.40	.22	1000	Packaging	.15	.15	.1	.05	1000	Demand	800	750	600	500		Unit profit	\$30	\$40	\$20	\$10		Unit penalty	\$15	\$20	\$10	\$8		[5]
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5.	Use the graphical method to solve the problem: Maximize $Z = 2x_1 + x_2$, subject to $x_2 \leq 10$ $2x_1 + 5x_2 \leq 60$ $x_1 + x_2 \leq 18$ $3x_1 + x_2 \leq 44$ and $x_1 \geq 0, \quad x_2 \geq 0.$	[5]																																																
6.	The following table summarizes the key facts about two products, A and B, and the resources, Q, R, and S, required to produce them.	[5]																																																

Resource	Resource Usage per Unit Produced		Amount of Resource Available
	Product A	Product B	
Q	2	1	2
R	1	2	2
S	3	3	4
Profit per unit	3	2	

All the assumptions of linear programming hold.

(a) Formulate a linear programming model for this problem.

(b) Solve this model graphically.

7. METRO WATER DISTRICT is an agency that administers water distribution in a large geographic region. The region is fairly arid, so the district must purchase and bring in water from outside the region. The sources of this imported water are the Colombo, Sacron, and Calorie rivers. The district then resells the water to users in the region. Its main customers are the water departments of the cities of Berdoo, Los Devils, San Go, and Hollyglass. [5]

It is possible to supply any of these cities with water brought in from any of the three rivers, with the exception that no provision has been made to supply Hollyglass with Calorie River water. However, because of the geographic layouts of the aqueducts and the cities in the region, the cost to the district of supplying water depends upon both the source of the water and the city being supplied. The variable cost per acre foot of water (in tens of dollars) for each combination of river and city is given in Table . Despite these variations, the price per acre foot charged by the district is independent of the source of the water and is the same for all cities.

The management of the district is now faced with the problem of how to allocate the available water during the upcoming summer season. In units of 1 million acre feet, the amounts available from the three rivers are given in the rightmost column of Table .

The district is committed to providing a certain minimum amount to meet the essential needs of each city (with the exception of San Go, which has an independent source of water), as shown in the *minimum needed* row of the table. The *requested* row indicates that Los Devils desires no more than the minimum amount, but that Berdoo would like to buy as much as 20 more, San Go would buy up to 30 more, and Hollyglass will take as much as it can get.

Management wishes to allocate *all* the available water from the three rivers to the four cities in such a way as to at least meet the essential needs of each city while minimizing the total cost to the district.

TABLE Water resources data for Metro Water District

	Cost (Tens of Dollars) per Acre Foot				Supply
	Berdoo	Los Devils	San Go	Hollyglass	
Colombo	16	13	22	17	50
River Sacron	14	13	19	15	60
River Calorie River	19	20	23	—	50
Minimum needed	30	70	0	10	(in units of 1 million acre feet)
Requested	50	70	30	∞	

- (a) Formulate this problem as a transportation problem by constructing the appropriate parameter table.
 (b) obtain an initial BF solution.

8. Briefly explain the characteristics of Operations Research. [5]
 Discuss the various steps used in solving Operations Research problems.

9. Four factories, A, B, C and D produce sugar and the capacity of each factory is given below: [5]
 Factory A produces 10 tons of sugar and B produces 8 tons of sugar, C produces 5 tons of sugar and that of D is 6 tons of sugar. The sugar has demand in three markets X, Y and Z. The demand of market X is 7 tons, that of market Y is 12 tons and the demand of market Z is 4 tons. The following matrix gives the returns the factory can get, by selling the sugar in each market. Formulate a transportation problem and solve for maximizing the returns.

	Profit in Rs. per ton ($\times 100$)			Availability in tons.
	Markets.			
Factories.	X	Y	Z	
A	4	3	2	10
B	5	6	1	8
C	6	4	3	5
D	3	5	4	6
Requirement in tons.	7	12	4	

10. You need to take a trip by car to another town that you have never visited before. Therefore, [5]
 you are studying a map to determine the shortest route to your destination. Depending on which route you choose, there are five other towns (call them A, B, C, D, E) that you might pass through on the way. The map shows the mileage along each road that directly connects two towns without any intervening towns. These numbers are summarized in the following table, where a dash indicates that there is no road directly connecting these two towns without going through any other towns.

Town	Miles between Adjacent Towns					Destination
	A	B	C	D	E	
Origin	40	60	50	—	—	—
A		10	—	70	—	—
B			20	55	40	—
C				—	50	—
D					10	60
E						80

Formulate this problem as a shortest-path problem by drawing a network where nodes represent towns, links represent roads, and numbers indicate the length of each link in miles.

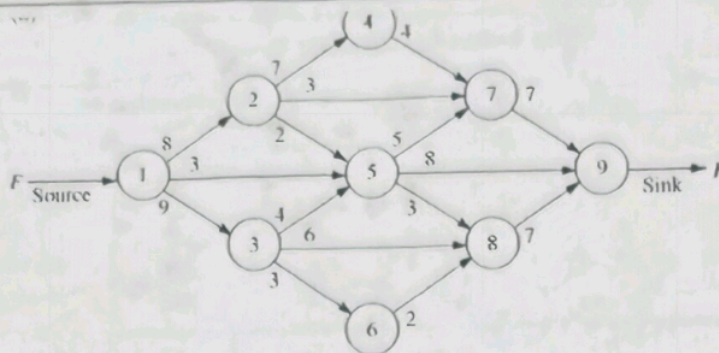
11. The Wirehouse Lumber Company will soon begin logging eight groves of trees in the same general area. Therefore, it must develop a system of dirt roads that makes each grove accessible from every other grove. The distance (in miles) between every pair of groves is as follows: [5]

	Distance between Pairs of Groves							
	1	2	3	4	5	6	7	8
1	—	1.3	2.1	0.9	0.7	1.8	2.0	1.5
2	1.3	—	0.9	1.8	1.2	2.6	2.3	1.1
3	2.1	0.9	—	2.6	1.7	2.3	1.9	1.0
4	0.9	1.8	2.6	—	0.7	1.6	1.5	0.9
5	0.7	1.2	1.7	0.7	—	0.9	1.1	0.8
6	1.8	2.6	2.3	1.6	0.9	—	0.6	1.0
7	2.0	2.3	1.9	1.5	1.1	0.6	—	0.5
8	1.5	1.1	1.0	0.9	0.8	1.0	0.5	—

Management now wishes to determine between which pairs of groves the roads should be constructed to connect all groves with a minimum total length of road.

- (a) Describe how this problem fits the network description of the minimum spanning tree problem.
 (b) Use the minimum spanning tree algorithm to solve the problem.

12. For a given networks, find the flow pattern giving the maximum flow from the source to the sink, given that the arc capacity from node i to node j is the number nearest node i along the arc between these nodes. [5]



13. The Dakota furniture company manufactures desks, tables and chairs. The manufacture of each type of furniture requires lumber and two types of skilled labour: finishing and carpentry. The amount of each resource needed to make each type of furniture is given in table below: [5]

Resource	Desk	Table	Chair
Lumber (board ft)	8	6	1
Finishing hours	4	2	1.5
Carpentry hours	2	1.5	0.5

Currently, 48 board feet of lumber, 20 finishing hour, and 8 carpentry hours are available. A desk sells for \$ 60, a table for \$ 30, and a chair for \$ 20. Dakota believes that demand for desks and chair is unlimited, but at most five tables can be sold. Formulate a LP model to maximize the total revenue of the company and solve it by Simplex Algorithm.

14. An auto company manufactures luxury cars and trucks. The company believes that its most likely customers are high-income women and men. To reach these groups, the company has embarked on an ambitious TV advertising campaign and has decided to purchase one minute commercial spots on two types of programs: comedy shows and football games. Each comedy commercial is seen by 7 million high income women and 2 million high income men. Each football commercial is seen by 2 million high income women and 12 million high income men. A one minute comedy ad costs Rs 50,000, and one minute football ad costs Rs 100,000. Auto company would like the commercial to be seen by at least 28 million high income women and 24 million high income men. Determine how auto company can meet its advertising requirements at minimum cost. [5]
